CIVL4340
Wind Engineering

Week 5: Atmospheric boundary layer flows

Overview

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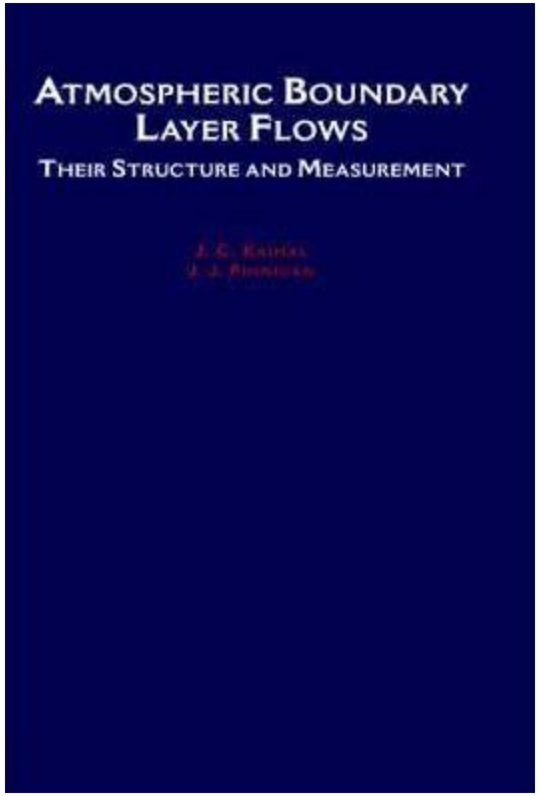
- Complete Week 4 lecture - covariance and correlation (postpone to next week)
- Turbulence spectra (postpone to next week)
- Non-homogeneous flows
    - Influence of topography
    - Non-homogeneous surface roughness
- Non-neutral stability
- AS/NZS1170.2 – Mzcat, Mh

Reading

- Holmes, Chapter 3
- Simiu and Yeo, Chapter 2
- *Kaimal and Finnigan 1994 textbook*



Available from library

Non-homogeneous flows

- Two primary cases of interest for wind engineering,
  - Influence of topography
  - Changes in surface roughness
- The influence of each depends on the structure of the ABL (what type of wind event is generating it).
- Mean and gust winds respond similarly, but at a different rate to each other.

http://www.hydro.com.au/home/

Topography

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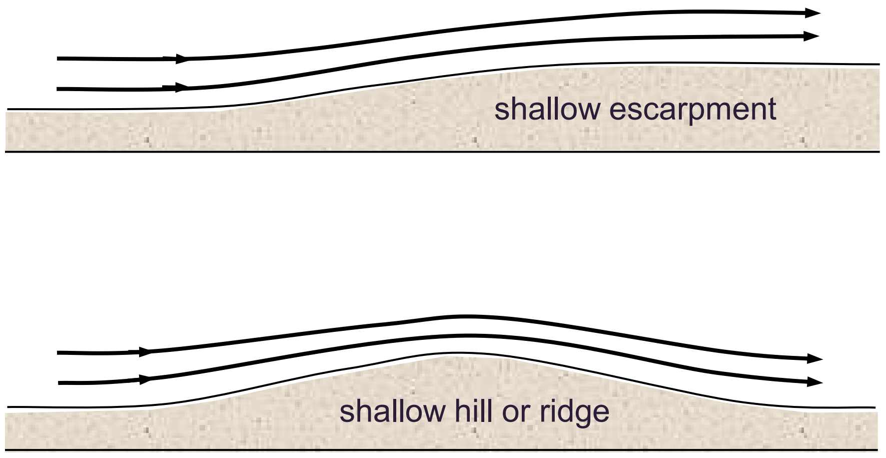
Shallow topography, $\phi < 0.3$



Holmes

$\phi = H/L$

$L = 2L_u$ in AS/NZS1170.2

- No flow separation (i.e. follows contours).
- Flow amplification everywhere (i.e. faster than without topo.) – peak at top of hill, ridge or escarpment.
- Predictable by computer models and wind tunnel tests.

Topography

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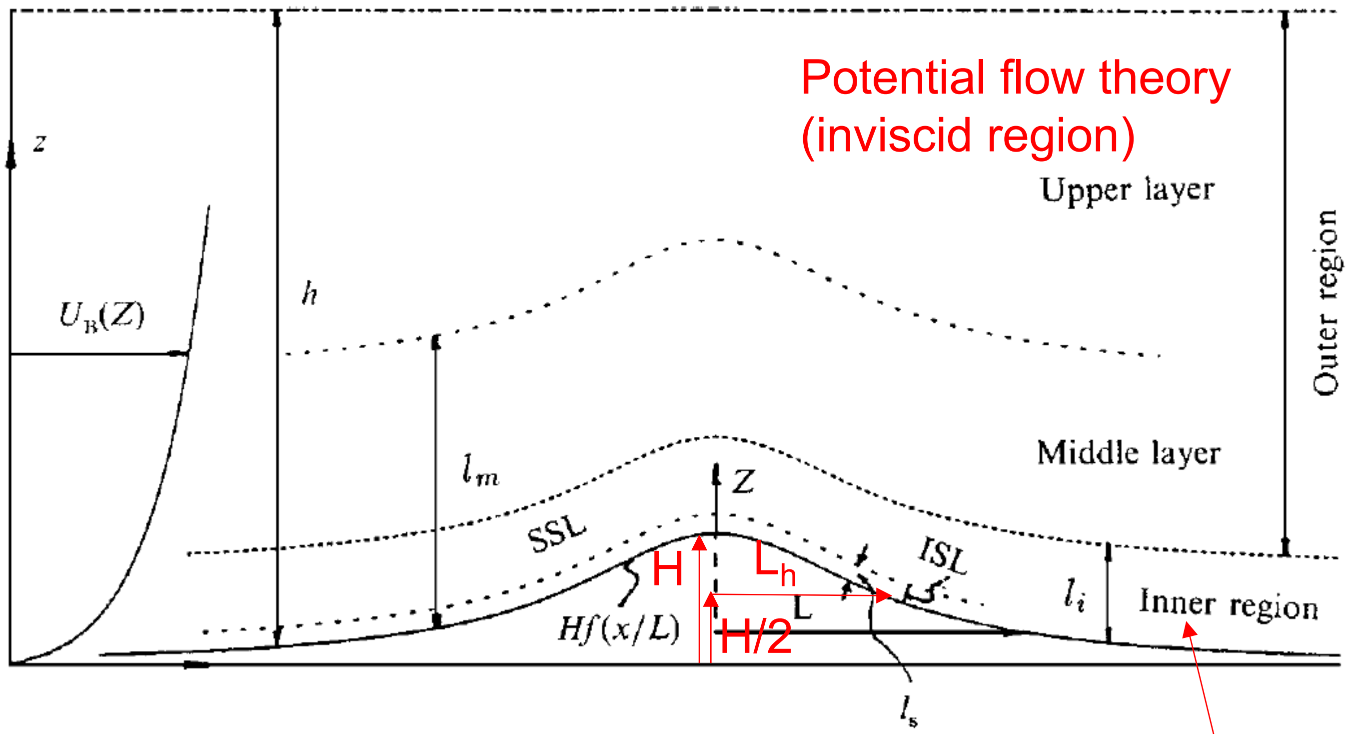
Shallow topography



$$l_m = L_h\left(ln\frac{L_h}{z_0}\right)^{-1/2}$$

Von Karman's constant, $\kappa = 0.4$

$$\frac{l_i}{L_h} ln\left(\frac{l_i}{z_0}\right) = 2\kappa^2$$

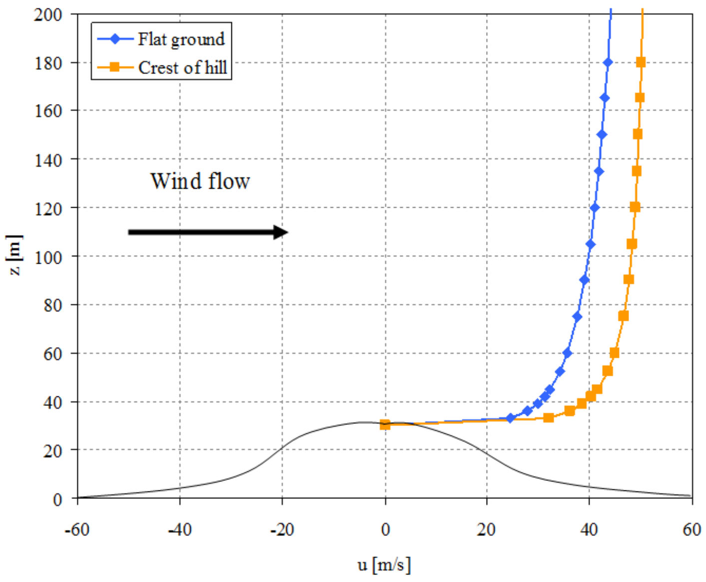
Analytical solutions
(linear theory)

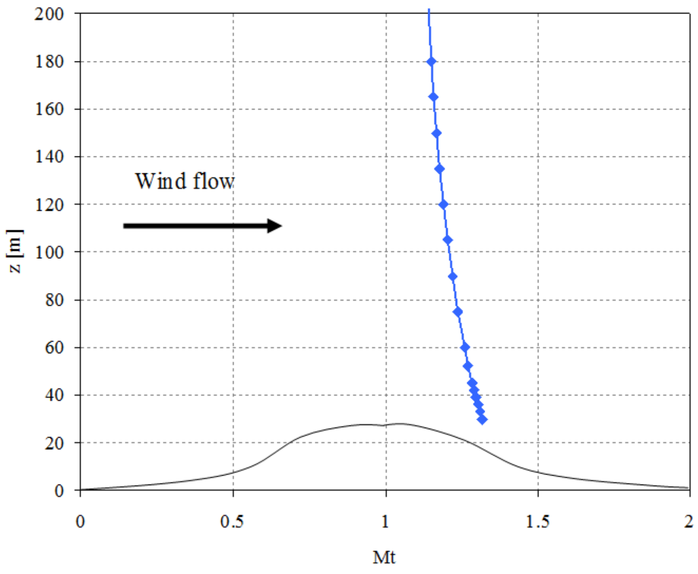
Example:
L_h = 200 m, z_0 = 0.02m
l_i = 10 m, l_m = 66 m

Topography

- Convenient to consider the influence of topography as a modification to the existing wind condition.
- Simply done by dividing the actual profile above a topographic feature with that for flat ground. This leads to the so-called, topographic multiplier, M_t.

$$M_t = \frac{Wind\ speed\ at\ height\ z\ above\ topography}{Wind\ speed\ at\ height\ z\ above\ flat\ ground\ upwind\ of\ site}$$





Topography

Mean flow over hill (Jackson & Hunt 1975)

$$\overline{M_t} = 1 + ks\phi$$

Shape constant

Position factor

Upwind slope

<u>Shape constant</u>
Ridge: 4
Escarpment: 1.6
3D hill: 3.2

s defines the shape of decay vertically and horizontally away from crest. *s* = 1 at the crest and is < 1 everywhere else.

Gust multiplier

$$\hat{M}_t = 1 + \hat{k}s\phi$$

$$\hat{k} = \frac{k}{1 + g\left(\frac{\sigma_u}{\overline{U}}\right)}$$

Topography

Derivation of gust multiplier (from Holmes)

$$\widehat{M_t} = \frac{\bar{U}\,\overline{M_t} + g\sigma_u}{\bar{U} + g\sigma_u} = \frac{\bar{U}\,(1 + ks\varphi) + g\sigma_u}{\bar{U} + g\sigma_u}$$

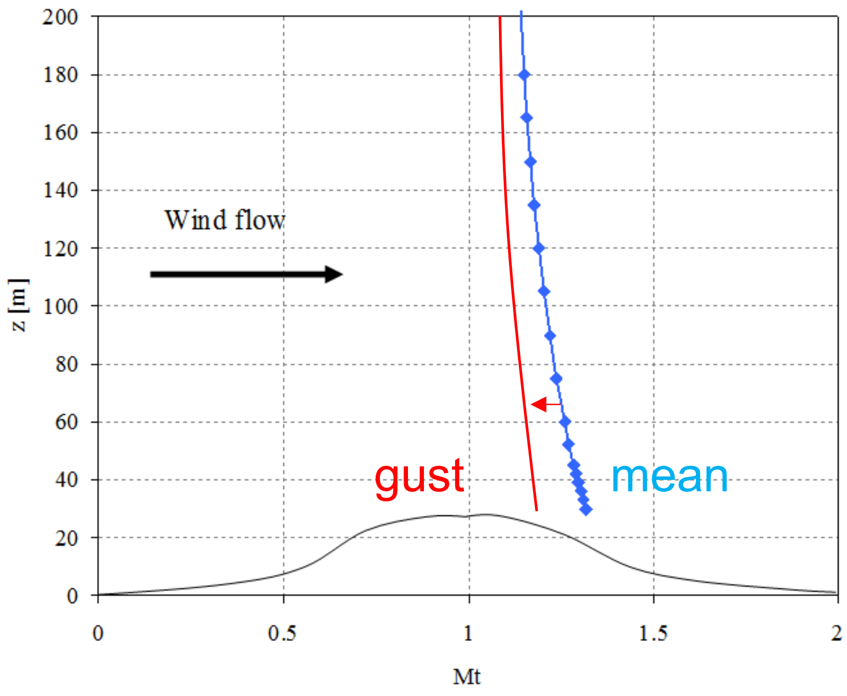
$$= \frac{(\bar{U} + g\sigma_u)\left(1 + \dfrac{ks\varphi}{1 + {g\sigma_u}/{\bar{U}}}\right)}{\bar{U} + g\sigma_u}$$

$$= \left(1 + \frac{ks\varphi}{1 + {g\sigma_u}/{\bar{U}}}\right)$$

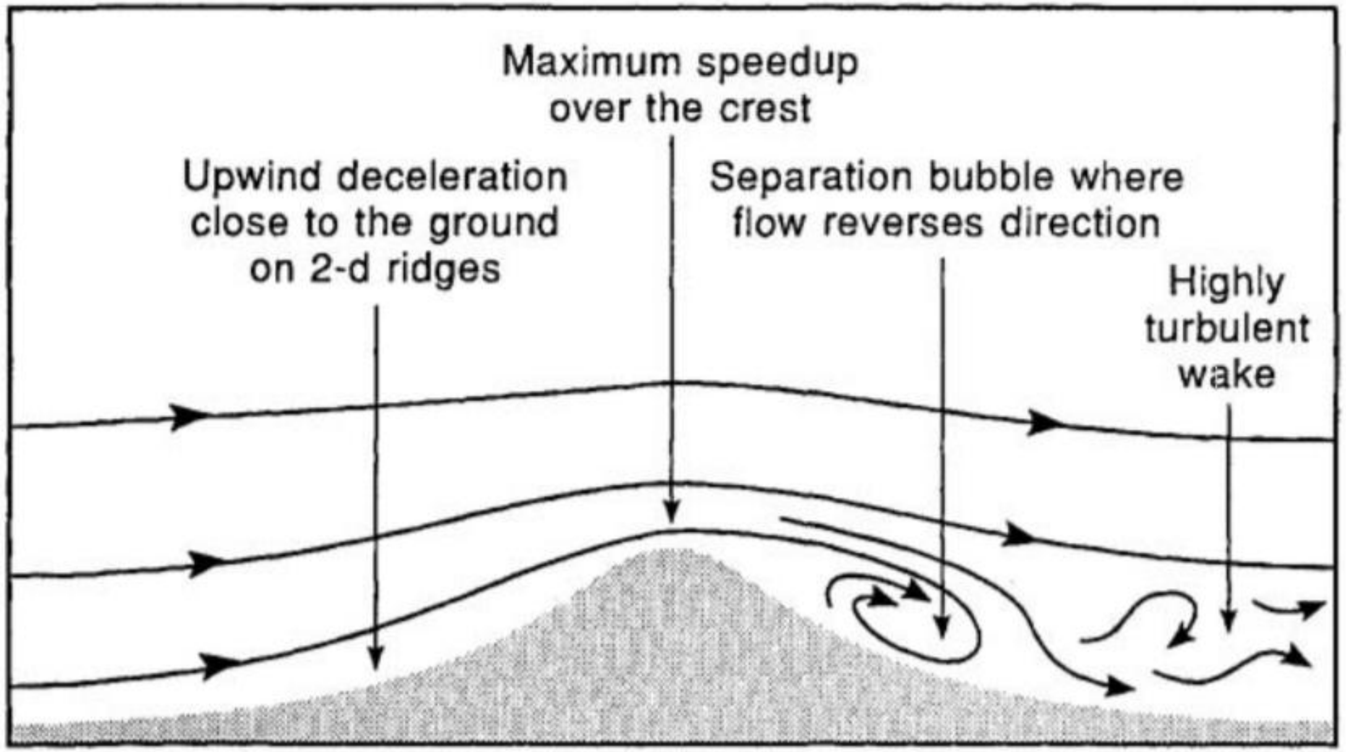
Which can be rewritten, $\widehat{M_t} = 1 + \hat{k}s\varphi$ where, $\hat{k} = \left(\dfrac{k}{1 + {g\sigma_u}/{\bar{U}}}\right)$

Topography

Result is that gust multipliers are less than mean

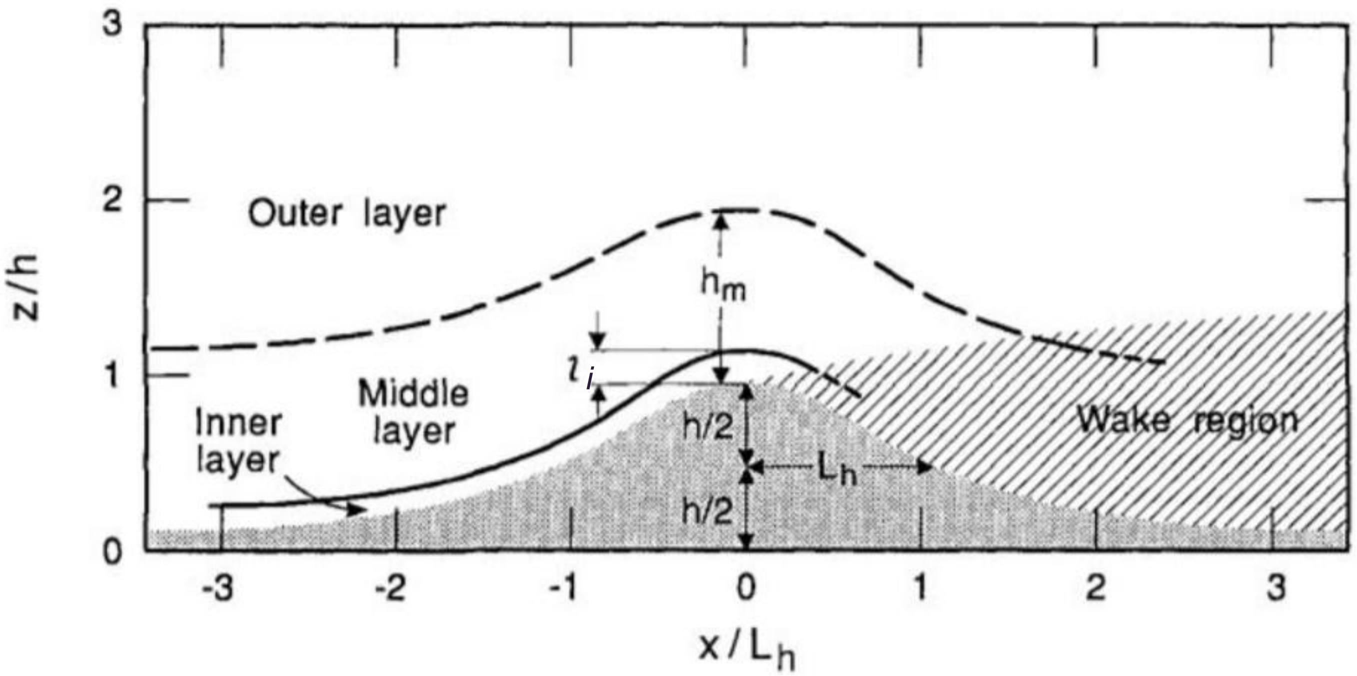


Steep topography



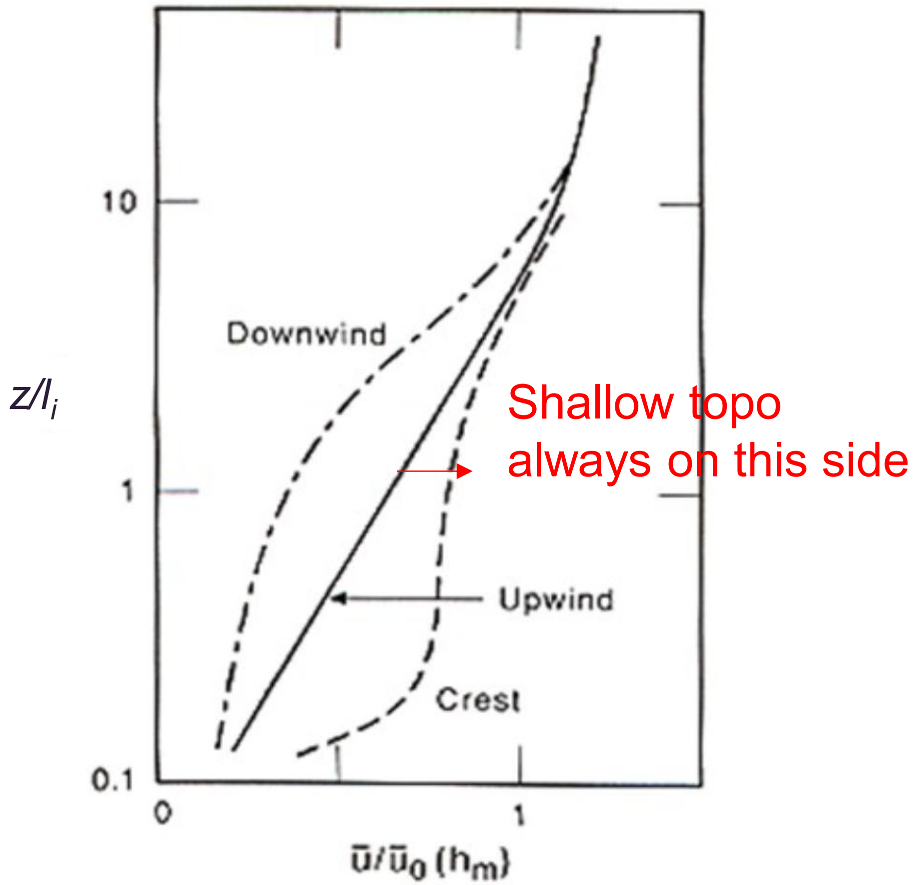
Kaimal and Finnigan (1994)

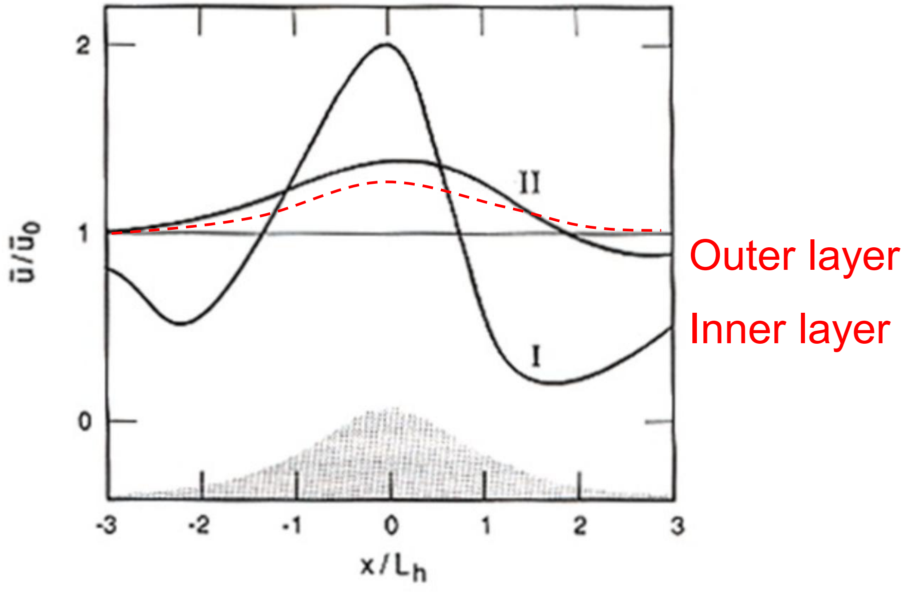
Steep topography



Kaimal and Finnigan (1994)

Steep topography

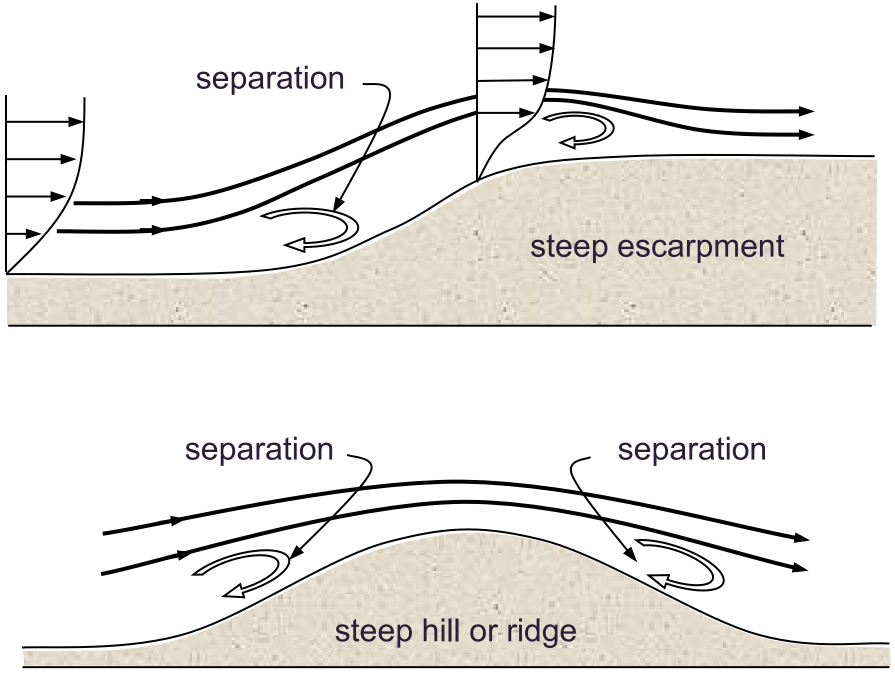




Kaimal and Finnigan (1994)

Steep topography

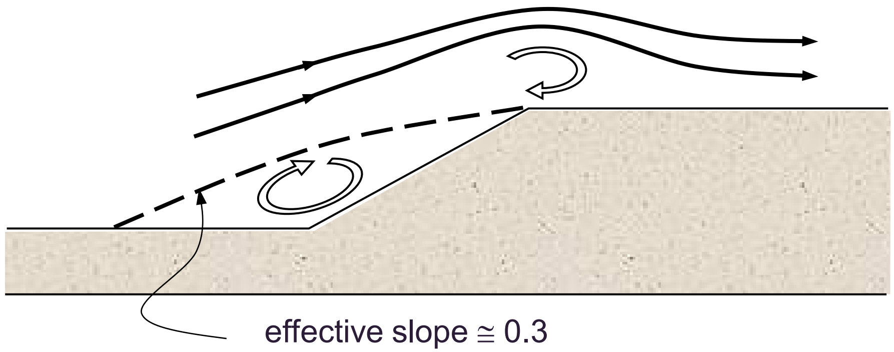
Steep topography, $0.3 < \phi < 1$



- Flow separation occurs.
- Flow reduction or reversal in places, large amplification in others.
- Numerical models struggle to predict influence.

Topography

Very steep topography, $\phi > 1$



Holmes

- Upper limit to amplification – recirculation zone creates an 'artificial' hill shape.
- Large topography 'sees' winds higher in the atmosphere, so not all increase in wind speed can be attributed to kinematic flow amplification.

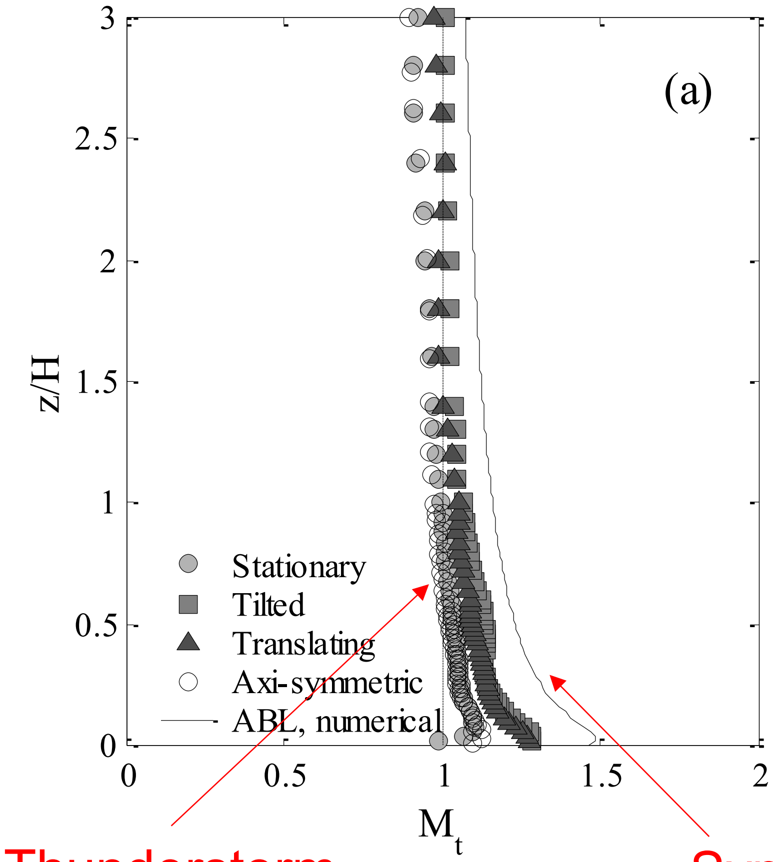
Topography

What about when we have non-synoptic wind events?

- When the topography is large compared to the boundary layer depth the topography can influence the storm.
- It may be expected that for TC winds with a lower gradient height the acceleration could be greater…..don't really know though.
- For thunderstorm winds acceleration is expected to be lower.

Topography

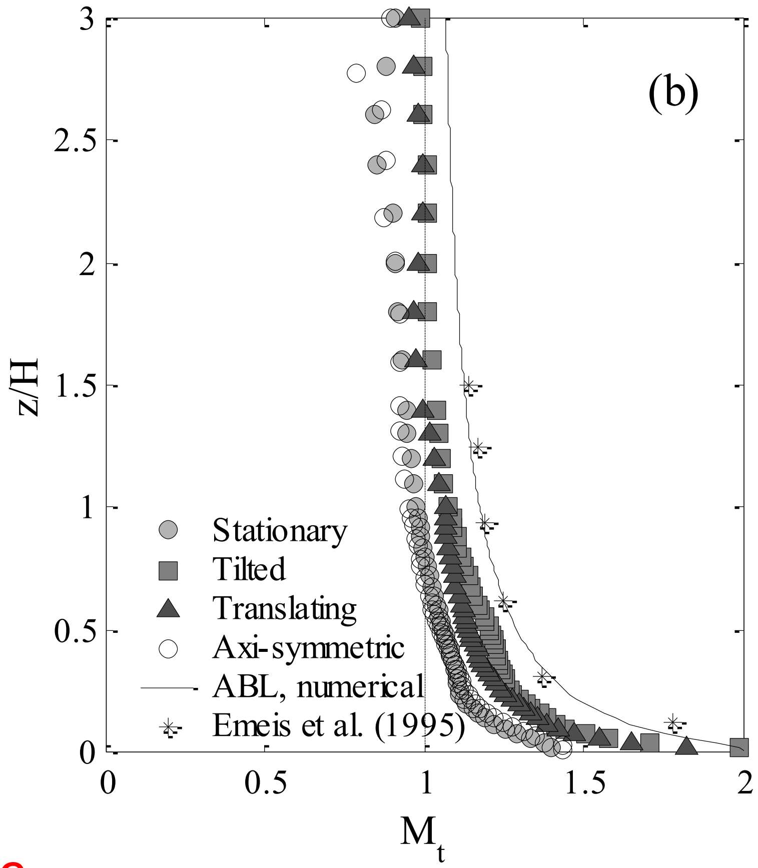
Φ = 0.2 escarpment



Thunderstorm

Synoptic

Φ = 0.5 escarpment



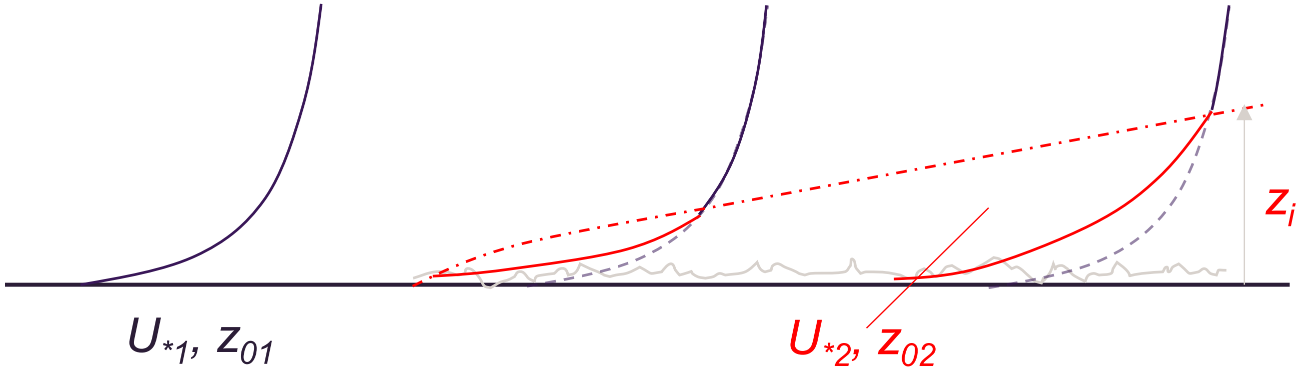
Topography

What about “real” topography?

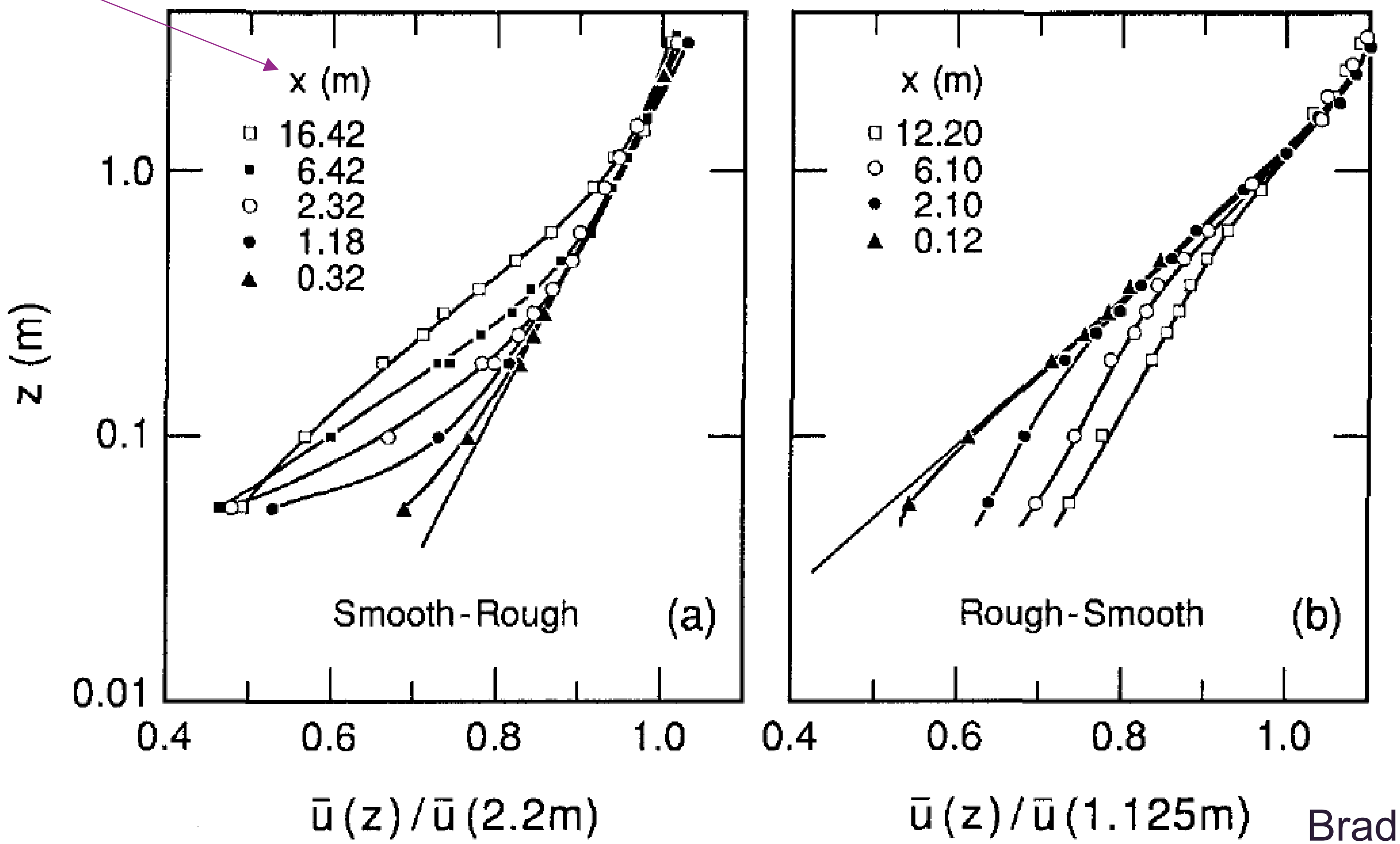
https://www.kenduncan.com/wp-content/uploads/2020/10/VX1404_l.jpg

Non-homogeneous terrain

- Most of the earth's surface is not flat, and not heterogeneous when it comes to surface conditions.
- Most winds are not in equilibrium with their underlying surface roughness throughout the entire ABL.
- This means the steady velocity profiles we usually apply (e.g. Log Law) may be inappropriate.
- Internal boundary layer (IBL) develops, which defines the new profile.



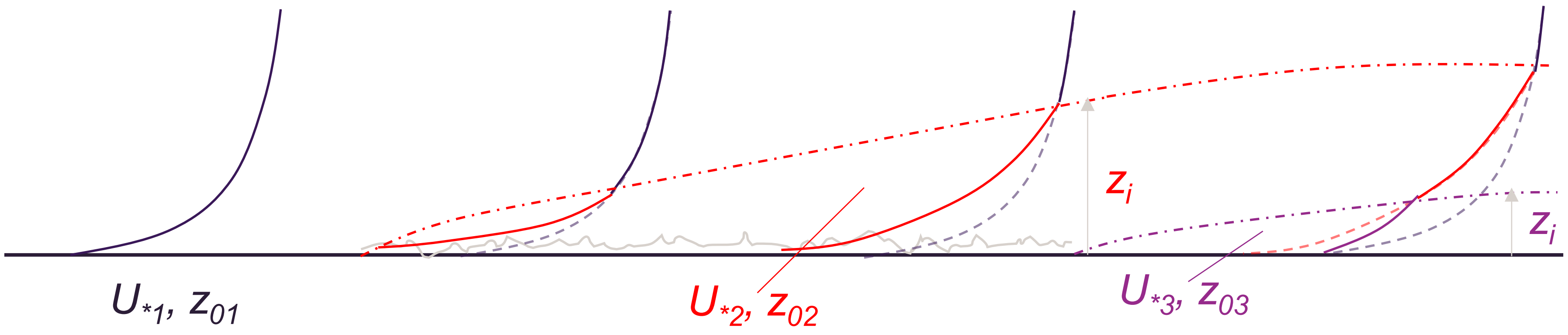
Distance from change in terrain



Bradley (1968)

Small-scale measurements

Non-homogeneous terrain



- **When multiple changes in terrain exist, multiple IBL form.**
- **Real boundary layers are very complicated!**
- **No simple way to define wind speed at a given *z* without IBL calculations – many models developed to estimate z_i.**

Non-homogeneous terrain

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- General ‘rules of thumb’ for practical application

1. A velocity profile can be assumed in equilibrium with its underlying surface conditions when distance to roughness change is > 5 km.

2. For distances < 500 m downwind of changes in roughness, the velocity profile can be assumed to be the same as upwind of the change. *(not always applied)*

3. Between 500 m and 5km, the velocity profile below z_i is logarithmic based on the underlying roughness downwind of the change, and constrained at z_i by the wind speed at that height based on the upwind profile.

Non-homogeneous terrain

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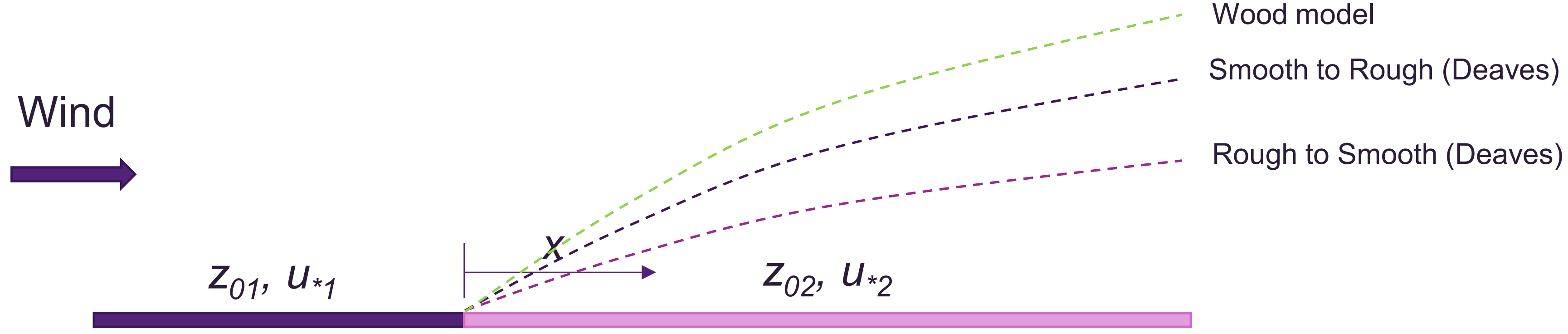
- Models for estimating z_i with a single roughness change.



Deaves model

z_{01} not considered

for $z_{01} < z_{02}$
Smooth to Rough

$$\frac{z_i}{z_{02}} = 0.36\left(\frac{x}{z_{02}}\right)^{0.75}$$

for $z_{01} > z_{02}$
Rough to smooth

$$\frac{z_i}{z_{02}} = 0.07\left(\frac{x}{\sqrt{z_{01}z_{02}}}\right)$$

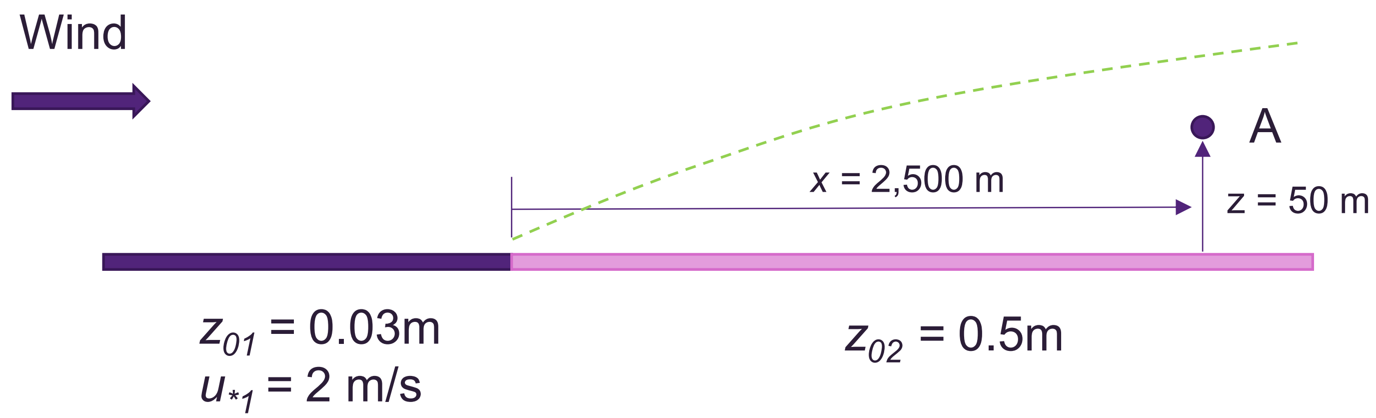
Wood model

$$\frac{z_i}{z_{0r}} = 0.28\left(\frac{x}{z_{0r}}\right)^{0.8}$$

Z_{0r} – greater of z_{01} and z_{02}.

Non-homogeneous terrain

- **Example:**
  - Determine the wind speed at A given the information below using the Wood Model of the IBL.

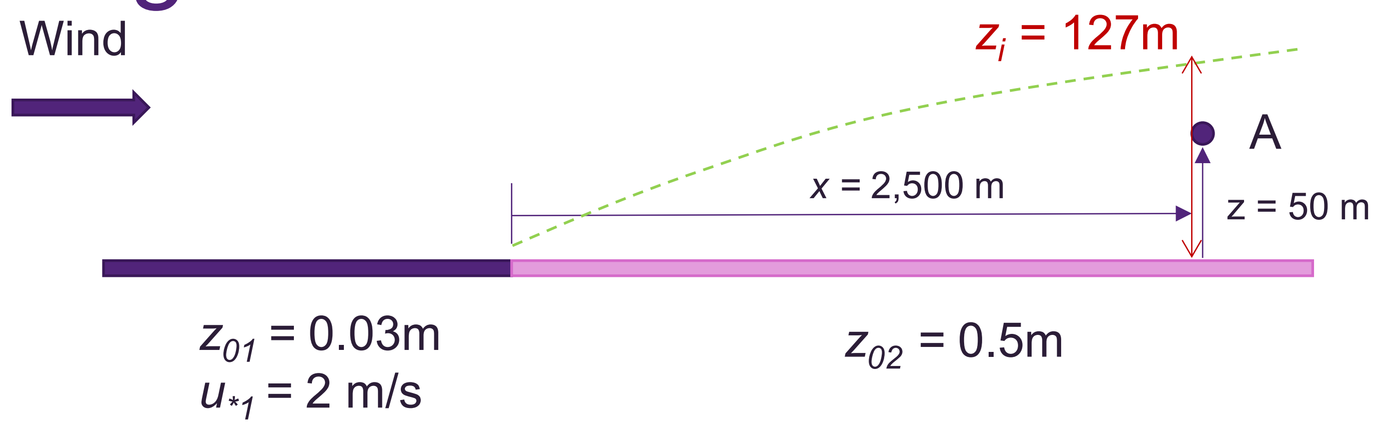


Calculate the IBL depth at A

$$\frac{z_i}{z_{0r}} = 0.28\left(\frac{x}{z_{0r}}\right)^{0.8}$$

$$z_i(x = 2{,}500) = 0.5 \times 0.28\left(\frac{2{,}500}{0.5}\right)^{0.8} = 127m$$

Non-homogeneous terrain



Using Log-Law

$$U(127m) = \frac{u_{*1}}{\kappa} ln\left(\frac{z_{iA}}{z_{01}}\right) = \frac{2}{0.4} ln\left(\frac{127}{0.03}\right) = 41.8m/s$$

Determine u_{*2}

$$u_{*2} = \frac{U(127m)\kappa}{ln\left(\frac{z_{iA}}{z_{02}}\right)} = \frac{41.8 \times 0.4}{ln\left(\frac{127}{0.5}\right)} = 3\ m/s$$

Therefore, calculate U(A)

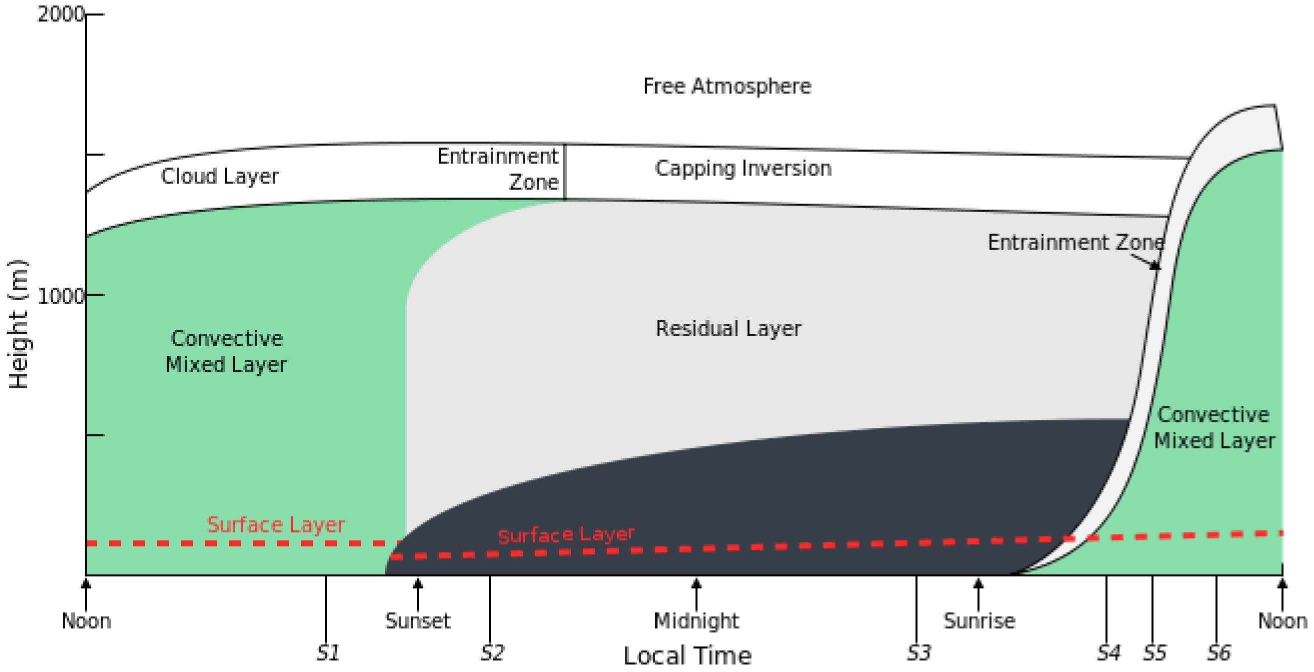
$$U(A) = \frac{3}{0.4} ln\left(\frac{50}{0.5}\right) = 34.5\ m/s$$

Non-homogeneous terrain

- Similar procedure followed for multiple changes in roughness – need to develop profiles through multiple IBLs.
- Conceptually, gust profile transition in a similar fashion to mean profiles discussed – but IBL growth is slower.
- AS/NZS1170.2 used to include a linearised version of a similar transition model, but now uses a simplified averaging model.

Non-neutral boundary Layers

- The general assumption in wind engineering is that strong winds are neutrally stratified – that is, the potential temperature remains constant through the boundary layer.
- Boundary layer stability (stratification) varies throughout the day, with implications for the structure of winds – of importance to wind energy prediction.
- Classified as unstable (convective), neutral or stable – principally based on a parameter known as the Obukhov Length.



https://en.wikipedia.org/wiki/File:Atmospheric_boundary_layer.svg

Non-neutral boundary Layers

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- Velocity profiles in non-neutral stability are governed by Monin-Obukhov Similarity Theory (MOST), which is a modified version of the standard Log Law

$$\overline{u}_H(z) = \frac{u_*}{\kappa}\left[\ln\left(\frac{z}{z_0}\right) - \Psi_1\left(\frac{z}{L}\right)\right]$$

- *L* is known as the Obukhov Length, and is defined as:

$$L = -\frac{u_*^3 \overline{T}}{\kappa g \overline{w'T'}},$$

Mean Temperature

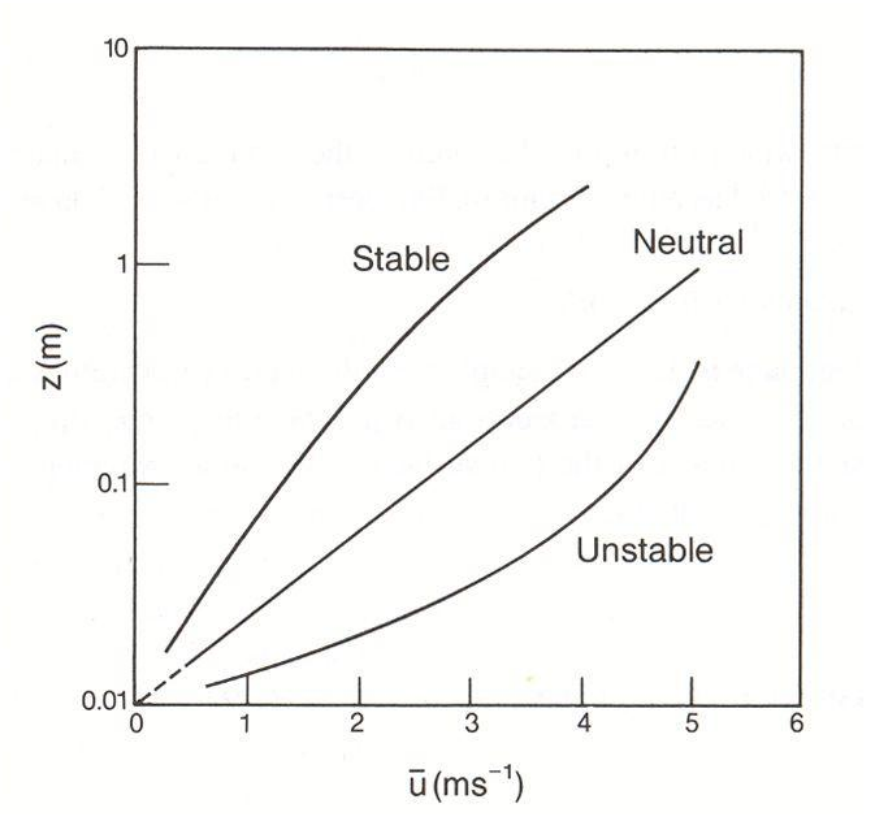
Vertical heat flux

- One approach for defining the stratification correction term $\psi_1(z/L)$, based on field experiments is:

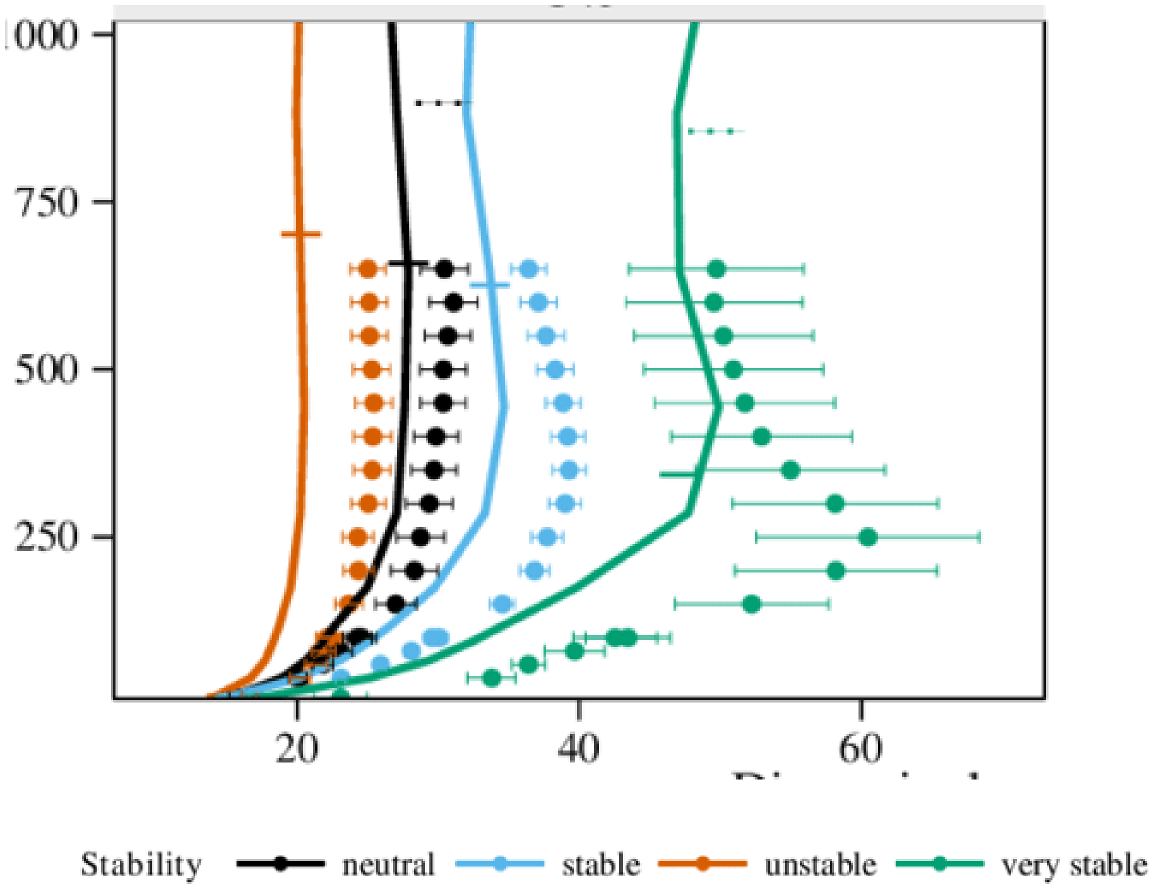
$$\Psi_1\left(\frac{z}{L}\right) = \begin{cases} 2\ln\left(\frac{1+x}{2}\right) + \ln\left(\frac{1+x^2}{2}\right) - \arctan(x) + \frac{\pi}{2}, & L \le 0 \quad \text{Unstable} \\ -4.7\frac{z}{L}, & L > 0, \quad \text{Stable} \end{cases}$$

$$x = \left(1 - 15\frac{z}{L}\right)^{1/4}.$$

Non-neutral boundary Layers



Wind profile in stable, neutral and unstable air.



Floors et al. (2013)

Non-neutral boundary Layers

- Stability also influences:
  - Turbulence characteristics
  - Topographic speed up

AS/NZS1170.2 – *a partial introduction*

Site wind speed ($V_{sit,\beta}$)

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$$V_{sit,\beta} = V_R M_c M_d (M_{z,cat} M_s M_t)$$

$V_{sit,\beta}$ depends on:

Region (V_R [3.2])

- This must also consider average recurrence interval determined based on Importance Level

Climate change multiplier (M_c [3.4]) (statistical)

Direction (M_d [3.3]) (statistical)

Terrain roughness & height ($M_{z,cat}$ [4.2])

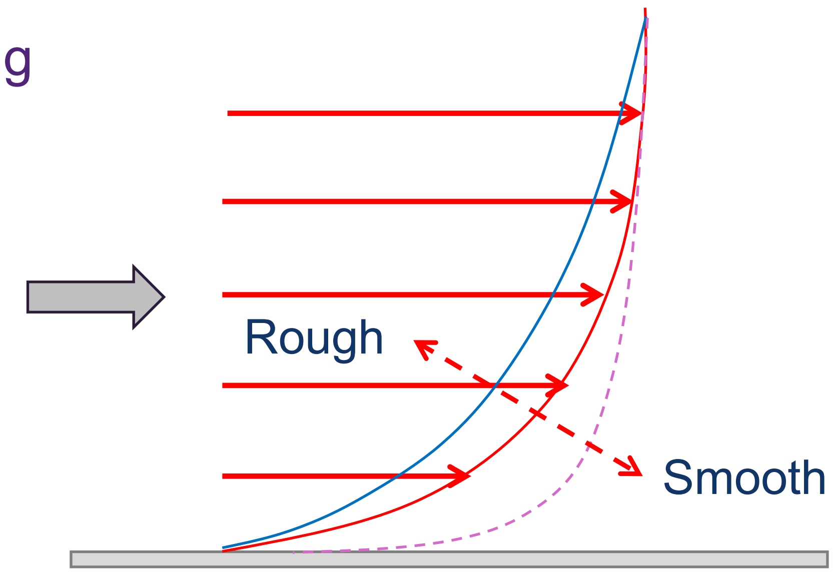
TODAY Shielding (M_s [4.3])

Topography (M_t [4.4]) (*M_{lee} considered if in NZ)*

Site wind speed ($V_{sit,\beta}$) must be calculated for 8 cardinal directions at reference height, h

Terrain/height multiplier ($M_{z,cat}$) [4.2]

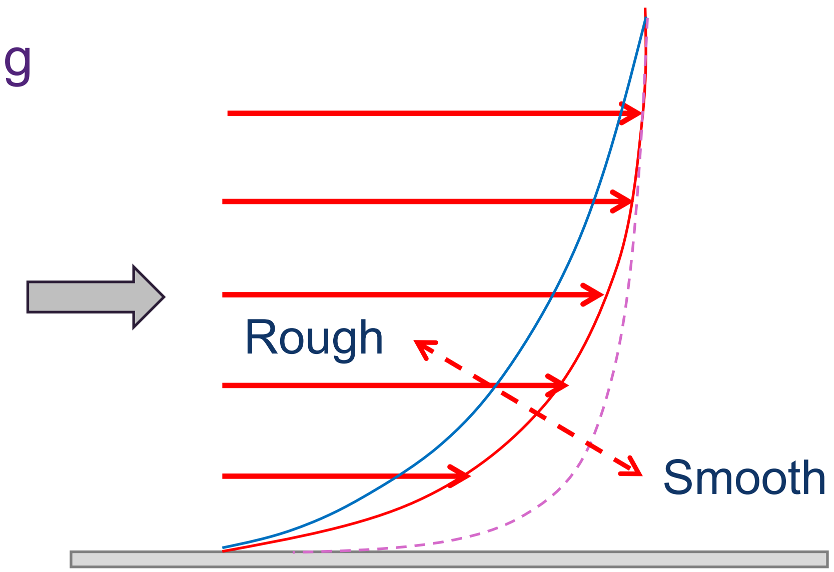
- Need to determine reference height (h)
- Consider terrain roughness upwind of building location (averaging generally required)
- $M_{z,cat}$ can be quite different for different directions



Terrain categories [4.2.1]

Terrain/height multiplier ($M_{z,cat}$) [4.2]

- Need to determine reference height (h)
- Consider terrain roughness upwind of building location (averaging generally required)
- $M_{z,cat}$ can be quite different for different directions



Example:
Location: Brisbane
Terrain Category 3
h = 20 m

$M_{z,cat}$ = 0.94

Averaging $M_{z,cat}$ [4.2.3]

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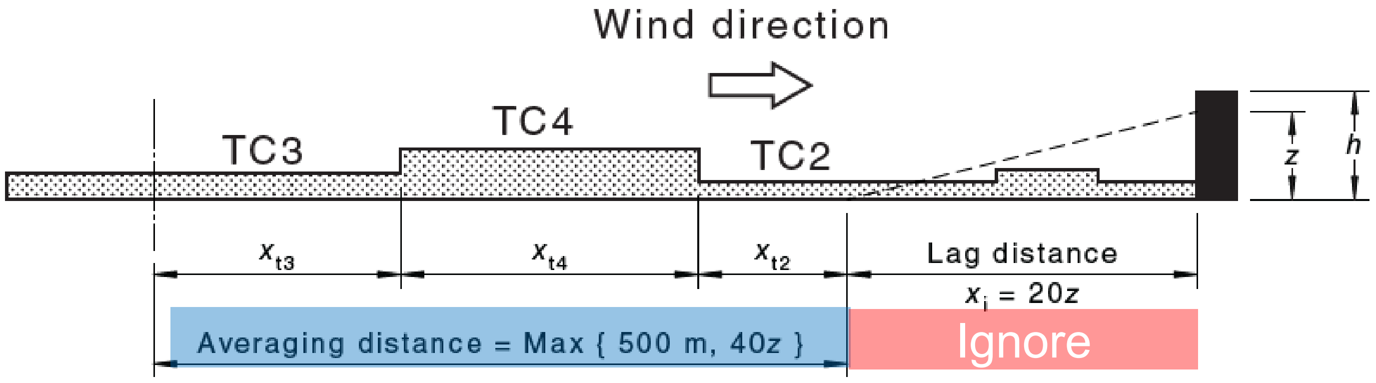
Averaging $M_{z,cat}$ [4.2.3]

- Weighted average over distance x_a = max(500 m, 40z)
- Ignore upwind distance, x_i = 20z



$$M_{z,cat} = \frac{M_{z,cat2}\, x_{t2} + M_{z,cat4}\, x_{t4} + M_{z,cat3}\, x_{t3}}{x_{t2} + x_{t4} + x_{t3}}$$ for the case illustrated

Averaging $M_{z,cat}$ [4.2.3]

Example:
z = 10 m
h = 20 m
$M_{z,cat}$: E & W

Lag distance:
$X_i = 20z$
$= 20 \times 10$ m
= 200 m

Average distance:
$X_a = \max(500\text{m}, 40z)$
= 500 m

Average $M_{z,cat}$:
$M_{z,cat\text{-}W} = (300 \times 1 + 200 \times 0.83)/500$
$= 0.93$
$M_{z,cat\text{-}E} = 0.83$

Topographic Multiplier (M_t) [4.4]

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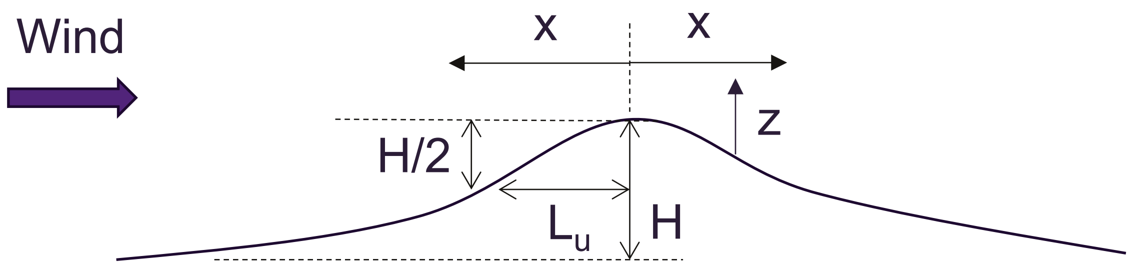
- Multiplier used to account for the influence of topography on wind speed.
- M_t depends on upwind slope and hill shape.
- Limited to 1.0 as a lower bound, so doesn't account for decrease in wind speed caused by topography (e.g. in valleys).
- Maximum value of 1.71.
- General equation:

$$M_h = 1 + \left(\frac{H}{3.5(z + L_1)}\right)\left(1 - \frac{|x|}{L_2}\right)$$



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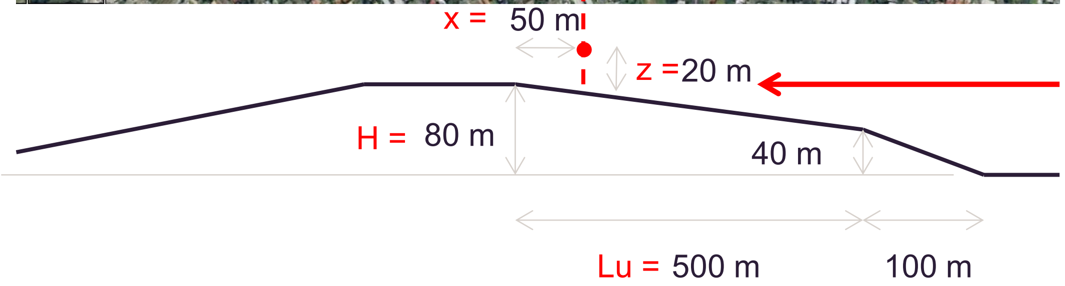
Topographic Multiplier (M_t) [4.4]



Assume z =20 m

$\phi = H/2L_u = 80/(500*2) = 0.08$

$\phi > 0.05$, intermediate slope

$$M_h = 1 + \left(\frac{H}{3.5(z+L_1)}\right)\left(1-\frac{|x|}{L_2}\right)$$

$L_1 = \max(0.36Lu, 0.4H) = 180m$
$L_2 = 4L_1 = 720m$

$M_t = M_h = 1.11$

Contact

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